

Daniel Marta

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RESEARCH STATEMENT

I am a postdoctoral researcher at ETH Zurich working on vision-language-action (VLA) models, reward models with large vision-language model (VLM) backbones, reinforcement learning, and large language models. I envision a world where robots and artificial agents interact safely and adapt seamlessly to humans through diverse feedback. My research focuses on aligning autonomous systems with human and environmental feedback: preference-based reinforcement learning for safe and sample-efficient robot adaptation [J1, C1–C5], and dense credit assignment from rich textual feedback for efficient LLM post-training [C6].

Keywords: Vision-Language-Action (VLA) Models, Reward Models, Vision-Language Models (VLMs), Reinforcement Learning (RL), Large Language Models (LLMs), RL from human feedback (RLHF)

SELECTED PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION, T=THESIS, *=EQUAL CONTRIBUTION

- [J1] **Marta, D.**, Pek, C., Melsión, G. I., Tumova, J., & Leite, I. (2021). Human-feedback shield synthesis for perceived safety in deep reinforcement learning. *IEEE Robotics and Automation Letters*, 7(1), 406-413.
- [C1] **Marta, D.**, Holk, S., Pek, C., Tumova, J., & Leite, I. (2023, May). Aligning human preferences with baseline objectives in reinforcement learning. In *2023 IEEE International Conference on Robotics and Automation (ICRA)* (pp. 7562-7568). IEEE.
- [C2] Holk, S., **Marta, D.**, Pek, C., & Leite, I. (2024, May). POLITE: Preferences combined with highlights in reinforcement learning. In *2024 IEEE International Conference on Robotics and Automation (ICRA)* (pp. 2288–2295). IEEE.
- [C3] Holk, S.*, **Marta, D.***, & Leite, I. (2024, March). PREDILECT: Preferences Delineated with Zero-Shot Language-based Reasoning in Reinforcement Learning. In *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction* (pp. 259-268).
- [C4] **Marta, D.***, Holk, S.*, Pek, C., & Leite, I. (2024, May). SEQUEL: Semi-Supervised Preference-based RL with Query Synthesis via Latent Interpolation. In *2024 IEEE International Conference on Robotics and Automation (ICRA)* (pp. 9585-9592). IEEE.
- [C5] **Marta, D.***, Holk, S.*, Vasco, M., Lundell, J., Homberger, T., Busch, F., Andersson, O., Kragic, D., & Leite, I. (2025). FLoRA: Sample-Efficient Preference-based RL via Low-Rank Style Adaptation of Reward Functions. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*.
- [C6] Hübötter, J., Lübeck, F., Behric, L. D., Baumann, A., Bagatella, M., **Marta, D.**, Hakimi, I., Shenfeld, I., Kleine Buening, T., Guestrin, C., & Krause, A. (2026). Reinforcement Learning via Self-Distillation. In *International Conference on Machine Learning (ICML)*.

EXPERIENCE

- **Postdoctoral Researcher** 2025 - Present
ETH Zurich Zurich, Switzerland
 - Conducting research on vision-language-action (VLA) models and reward models with large vision-language model (VLM) backbones.
 - Developing reinforcement learning and large language model methods for robotic and agentic systems.
- **Visiting Researcher** 2024 - 2024
Interactive Machines Group, Computer Science Department, Yale University United States
 - Explored enhancing interpretability of human feedback when modeling reward functions in the context of RLHF.
- **Research Engineer** 2018 - 2018
RL Team, Waymo Research United Kingdom
 - Implemented distributed RL agent learning in Docker containers for self-driving car simulations.
- **Research Intern** 2017 - 2018
RL Team, Waymo Research United Kingdom
 - Implemented RL algorithms such as DDPG and DQN.
- **Graduate Research Fellow** 2016 - 2017
GAIPS, INESC-ID Portugal
 - Researched unsupervised learning methods within RL contexts.

EDUCATION

- **PhD in Computer Science** 2020 - 2025
RPL (Robotics, Perception and Learning) Department, KTH Royal Institute of Technology Stockholm, Sweden
 - Completed PhD in 2025.
 - Enhanced Preference-based RL methods for robot control tasks which can adapt to human feedback.
 - Devised RLHF methods to enhance query information for human-robot interaction.
 - Supervisors: Prof. Iolanda Leite, Prof. Jana Tumova.
- **Doctoral Researcher** 2019 - 2020
ATO (Air Transport and Operations) Department Delft, Netherlands
 - Investigated combinatorial optimization in RL to predict aircraft scheduling and passenger preferences for air transport operations.
- **B.Sc. & M.Sc. in Aerospace Engineering, Avionics (Grade: 16/20)** 2009 - 2016
Technical University of Lisbon Lisbon, Portugal
 - Thesis: *Deep Learning Methods for Reinforcement Learning*.
 - Supervisor: Prof. Rodrigo Ventura.

HONORS AND AWARDS

- **Best Paper Awards**
 - Best student paper award for an outstanding contribution to the WACV 2024 LLVM-AD workshop.
- **Nominations**
 - Paper C2 was a finalist for the Best HRI Paper Award, Best Student Paper Award, and Best Paper Award at the 2024 IEEE International Conference on Robotics and Automation (ICRA).
- **Grants**
 - Awarded a PhD Grant for the WASP doctoral program from the Wallenberg foundation, which is the largest individual research program in Sweden.
 - Awarded a Research Exchange Grant, to visit Yale University in the spring of 2024.

TEACHING AND SUPERVISION

- **Teaching** 2020 - 2024
DD2380 AI course
 - Teaching assistant and leader of the RL assignment; lectured RL tutorials and coordinated teams of teaching assistants to deliver lectures and in-person examinations.
 - Developed and maintained the RL assignment and the dockerized evaluation process.
- **Supervision**
 - Oskar Stigland, Year:2024, Thesis: Explainability and Hierarchical Reinforcement. Learning
 - Mohammed Akif, Year:2023, Thesis: Safe Reinforcement Learning for Social Human-Robot Interaction

SKILLS

- **Programming & Scripting Languages:** Proficient in Python (socket networking, multi-processing/threading), Moderate experience with C++, Javascript, SQL, XML/XSL, $\text{\LaTeX}$.
- **Machine Learning Libraries:** Proficient in PyTorch (current research focus) and TensorFlow (distributed learning in previous research engineer role).
- **Simulation Engines:** Implemented several RL environments in Unity, familiar with socket APIs for Python, moderate experience in Unreal Engine.
- **Containerization & Deployment:** Proficient in Docker, moderate experience in Kubernetes.
- **Web Development Frameworks:** Moderate experience with Angular.
- **Other Tools & Technologies:** Moderate experience in SolidWorks, minor experience with Blender and Photoshop.

ACADEMIC SERVICE

- **IEEE International Conference on Robotics and Automation (ICRA)** 2020 - 2024
Reviewer
- **IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)** 2022, 2023
Reviewer
- **IEEE Robotics and Automation Letters** 2022, 2023
Reviewer
- **ACM/IEEE International Conference on Human-Robot Interaction (HRI)** 2023, 2024
Reviewer